

$$\begin{array}{c}
\begin{array}{c}
T^2[TX, Y] \xrightarrow{\mu[TX, Y]} T[TX, Y] \xrightarrow{\xi_{TX, Y}} [T^2X, TY] \xrightarrow{[\eta_{TX, TY}]} [TX, TY] \\
\downarrow T^2[\mu_X, Y] \quad \searrow T\xi_{TX, Y} \quad (20) \quad \downarrow [\mu_{TX, TY}] \quad \text{monad} \quad \equiv \quad \downarrow [\eta_X, TY] \\
T^2[T^2X, Y] \xrightarrow{\text{nat } T\xi} T[T^2X, TY] \xrightarrow{[\eta_{T^2X, TY}]} [T^2X, TY] \xrightarrow{[\eta_{TX, TY}]} [TX, TY] \xrightarrow{[\eta_X, TY]} [X, TY] \xleftarrow{[X, \mu_Y]} \\
\downarrow T\xi_{T^2X, Y} \quad \nwarrow T[T\mu_X, TY] \quad \searrow \xi_{T^2X, TY} \quad \uparrow [T^3X, \mu_Y] \quad \equiv \quad \uparrow [X, \mu_Y] \\
T[T^3X, TY] \xrightarrow{[T\eta_{T^2X, TY}]} T[T^2X, TY] \xrightarrow{[\eta_{TX, TY}]} [T^2X, TY] \xrightarrow{[\eta_{TX, T^2Y}]} [TX, T^2Y] \xrightarrow{[\eta_X, T^2Y]} [X, T^2Y] \xleftarrow{[\eta_X, T^2Y]} \\
\downarrow T[\eta_{T^2X, TY}] \quad \text{monad} \quad \searrow \text{nat } \xi \quad \downarrow [T\eta_{TX, T^2Y}] \quad \text{nat } \eta \\
T[T^2X, TY] \xrightarrow{T[\eta_{TX, TY}]} T[TX, TY] \xrightarrow{\xi_{TX, TY}} [T^2X, T^2Y] \xrightarrow{[\eta_{TX, T^2Y}]} [TX, T^2Y]
\end{array}
\end{array}$$